

Saumyen Kundu Ph.D.

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Date of Birth: 13th July, 1991

Profile

Motivated and hard-working young academic with a Ph.D. in Theoretical Physics and significant experience in classroom-teaching across undergraduate and postgraduate levels. My research in BSM physics, dark matter, and collider phenomenology with publications in reputed international journal enhances my classroom instruction with contemporary scientific context. Committed to strengthen students' analytical abilities and overall academic progress through a supportive and collaborative teaching environment.

Work Experience

Apr, 2024– present	<i>Post-Doctoral Fellow</i> Harish-Chandra Research Institute Prayagraj, Uttar Pradesh, India
Sep, 2015– May, 2017	<i>Assistant Professor</i> , Department of Physics Seacom Skills University, Sriniketan, West Bengal

Teaching Experience

Jan, 2025 – May, 2025	<i>Teaching Assistant</i> Harish-Chandra Research Institute, Prayagraj, Uttar Pradesh <i>Particle Physics</i> course for Ph.D. and M.Sc. students.
Aug 2017 – Dec 2022	<i>Teaching Assistant</i> BITS Pilani, K.K. Birla Goa Campus, Zuarinagar, Goa <i>Particle Physics</i> course for M.Sc. students. <i>Laboratory Physics</i> course for B.E. students.
Sep 2015 – May 2017	<i>Assistant Professor</i> Seacom Skills University, Sriniketan, West Bengal <i>Physics (Honours)</i> for undergraduate students. <i>Engineering Physics</i> and <i>Electromagnetic Field Theory</i> for B.Tech. students.

Education

2017 – 2023	Ph.D. in High Energy Physics Advisor: Prof. Prasanta Kumar Das BITS Pilani, K.K. Birla Goa Campus , Goa
2013 – 2015	M.Sc. in Physics (specialization in <i>Particle Physics</i>) Visva-Bharati , Santiniketan, West Bengal
2010 – 2013	B.Sc. in Physics (Honours) Visva-Bharati , Santiniketan, West Bengal

Technical Skills

- **Software packages:** CERN-ROOT, TMVA, Mathematica, FeynRules, SARAH, SPHeno, LanHEP, MadGraph5, Whizard, CalcHEP, Pythia8, Delphes3, FastJet3, MadDM, MicrOMEGAs, MadAnalysis5
- **Languges:** C++, Python, bash-script.
- **Python Libraries:** NumPy, SciPy, Pandas, Matplotlib, Seaborn, XGBoost.
- **Other:** Microsoft Office, Google Docs, Sheets, Slides, L^AT_EX.

Research Interests

- Beyond Standard Model (BSM) Phenomenology
- Dark Matter Search
- Collider Phenomenology
- Higgs Physics
- Applications of Machine Learning Algorithms

Publications

For the *Articles*, Authors are listed alphabetically by the Last Names.

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- Articles*
- *Tracing Neutrino Non-Standard Interactions through Charged Lepton Collisions*
Sudip Jana, **Saumyen Kundu**, Santosh Kumar Rai
under review in Physics Letters B [[arXiv:2507.10703](#)].
 - *Revisiting the inert scalar dark matter with vector-like quarks*
Prasanta Kumar Das, Shyamashish Dey, **Saumyen Kundu**, Santosh Kumar Rai
JHEP **05** (2025) 027 [[arXiv:2412.17719](#)].
 - *Exploring the electromagnetically interacting dark matter at the International Linear Collider*
Manish Kumar Sharma, **Saumyen Kundu**, Prasanta Kumar Das
under review in Physics Letters B [[arXiv:2301.02514](#)].
 - *Jet substructure probe to unfold singlet-doublet dark matter in the presence of non-standard cosmology*
Prasanta Kumar Das, Partha Konar, **Saumyen Kundu**, Sudipta Show
JHEP **06** (2023) 198 [[arXiv:2301.02514](#)].
 - *SN1987A cooling due to Plasmon-Plasmon scattering in the Randall-Sundrum Model*
Manish Sharma, **Saumyen Kundu**, Prasanta Kumar Das
Phys. Dark Univ. **40** (2023) 101218 [[arXiv:2304.00380](#)].
 - *EFT analysis of leptophilic dark matter at future electron-positron colliders in the mono-photon and mono-Z channels*
Saumyen Kundu, Atanu Guha, Prasanta Kumar Das, P.S. Bhupal Dev
Phys. Rev. D **107** (2023) 015003 [[arXiv:2110.06903](#)].
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- Conference Proceedings*
- *Enhancing Dark Matter Phenomenology with Vector-Like Quarks in the IDM*
Prasanta Kumar Das, Shyamashish Dey, **Saumyen Kundu**, Santosh Kumar Rai
Springer Proc. Phys. **432** (2026) 437-440.
Contribution to: [26th DAE-BRNS High Energy Physics Symposium](#).
 - *Jet substructure probe to freeze-in dark matter in alternative cosmological background*
Saumyen Kundu, Sudipta Show, Partha Konar, Prasanta Kumar Das
Eur. Phys. J. Spec. Top. **233**, 2129–2134 (2024).
 - *Freeze-In Dark Matter Search at Colliders with Jet Substructure Analysis*
Saumyen Kundu, Sudipta Show, Partha Konar, Prasanta Kumar Das
Springer Proc. Phys. **304** (2024) 393-396.
Contribution to: [25th DAE-BRNS High Energy Physics Symposium](#).
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- Reports*
- *The International Linear Collider: Report to Snowmass 2021*
ILC International Development Team
Contribution to: [Snowmass 2021](#) e-Print: [2203.07622 \[physics.acc-ph\]](#).

Academic & Professional Service

- Reviewer, Indian Journal of Physics (InJP).
- Reviewer, International Journal of High Energy Physics (IJHEP)).
- Member, Organizing Team, *Sangam@HRI 2026* – An Instructional Workshop in Particle Physics.
- Member, Organizing Team, *Pravaha 2026* – HRI HEP Meeting.
- Developer & Maintainer, Website of High Energy Physics Group at Harish-Chandra Research Institute.

Recent Talks and Presentations

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| 19 – 23 Dec,
2024 | <i>Enhancing Dark Matter Phenomenology with Vector-Like Quarks in the IDM</i>
XXVI DAE-BRNS HEP Symposium , Banaras Hindu University, India
Poster Presentation |
| 10 – 12 March,
2023 | <i>Jet substructure probe to freeze-in dark matter in alternative cosmological background</i>
Frontiers in Particle Physics, 2023 , CHEP, IISc, Bangalore, India
Poster Presentation |
| 12 December,
2022 | <i>Freeze-in Dark Matter Search at Colliders with Jet Substructure Analysis</i>
XXV DAE-BRNS HEP Symposium , IISER Mohali, India
Oral Presentation |
| 07 December,
2021 | <i>Leptophilic Dark Matter at International Linear Collider</i>
AstroDark-2021 , Kavli IMPU, Japan
Poster Presentation |
| 11 November,
2021 | <i>Searching for Dark Matter signatures at the ILC with Leptophilic Effective Models</i>
Anomalies 2021 , IIT Hyderabad, India
Oral Talk |
| 04 November,
2021 | <i>Searching for Leptophilic Dark Matter in Linear Colliders</i>
Brookhaven Forum 2021 , Brookhaven National Laboratory, USA
Oral Talk |
| 27 October,
2021 | <i>Quest for Dark Matter at International Linear Collider</i>
ILC Workshop on Potential Experiments (ILCX2021)
Oral Talk |
| 25 January,
2020 | <i>Searching for Leptophilic Dark Matter in Linear Colliders</i>
Goa Physics Meet 2020 , IIT Goa, Ponda, Goa
Poster Presentation |

Schools/Workshops/Conferences Attended

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| 14 November–
18 November, 2021 | • <i>17th International Conference on Interconnections between Particle Physics and Cosmology (PPC 2025)</i>
IIT Hyderabad, India |
| 13 March–
24 March, 2021 | • <i>Less Travelled Path to the Dark Universe, 2023</i>
TIFR-ICTS, Bengaluru, India |
| 29 November–
2 December, 2021 | • <i>Kashiwa Dark Matter Symposium, 2021</i>
ICRR, University of Tokyo, Japan
Online |
| 22 November–
26 November, 2021 | • <i>Computational Tools for High Energy Physics and Cosmology</i>
Institut de Physique des 2 Infinis de Lyon (IP2I), Lyon, France
Online |

- 22 November–
26 November, 2021 • *Hunting SUSY @ HL-LHC, 2021*
ICTS-TIFR, Bengaluru, India
Online
- 18 October–
22 October, 2021 • *Higgs 2021*
Stony Brook University, Simons Center for Geometry and Physics
Online
- 23 August–
4 September, 2021 • *CERN-Fermilab Hadron Collider Physics Summer School, 2021*
CERN, Switzerland
Online
- 28 July–
6 August, 2020 • *International Conference on High Energy Physics, 2020*
Prague, Virtual Conference
- 18 – 22 November,
2019 • *MadGraph School 2019*
Institute of Mathematical Science, Chennai, Tamil Nadu
- 14 October–,
9 November, 2019 • *SERB Preparatory School in Theoretical High Energy Physics*
Tezpur University, Tezpur, Assam
- 3 – 5 February,
2019 • International Conference on “*Recent Issues in Nuclear and Particle Physics*”
Visva-Bharati, Santiniketan, West Bengal

References

- *Prof. Prasanta Kumar Das*
Professor
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BITS Pilani, K.K. Birla Goa Campus, Goa, India
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 - *Prof. Santosh Kumar Rai*
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 - *Prof. Partha Konar*
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Theoretical Physics Division,
Physical Research Laboratory, Ahmedabad, India
konar@prl.res.in
 - *Prof. P.S. Bhupal Dev*
Associate Professor
Department of Physics and McDonnell Center for the Space Sciences,
Washington University, St. Louis, Missouri, USA
bdev@wustl.edu
 - *Dr. Sudip Jana*
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